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Electronic apparatus and command accessing method thereof having power saving mechanism

Abstract

The present disclosure discloses a command accessing method having power saving mechanism is provided that includes steps outlined below. A cache control circuit is controlled to retrieve a plurality of commands from a command source to be executed. A plurality of loop commands corresponding to a loop procedure included in the commands are determined, wherein the loop commands include a first part that matches a storage amount of a command buffer circuit and a second part that exceeds the storage amount. The cache control circuit is controlled to store the first part of the loop commands to the command buffer circuit and not store the second part of the loop commands to the command buffer circuit. The cache control circuit is controlled to, every time the loop procedure is performed, retrieve and execute the first part of the loop commands from the command buffer circuit and retrieve and execute the second part of the loop commands from the command source.

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Summary of the TW OA letter: 1. Claims 1-10 are rejected as allegedly being unpatentable in view of cited reference 1 (TW 201612744 A, also published as US20160092230A1). Correspondence between claims of TW counterpart application and claims of US application: 1. Claims 1, 3, . . . , 6 and 8-10 in TW counterpart application correspond to claims 1, 2-3, . . . and 8 in US application, respectively. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to an electronic apparatus and a command accessing method thereof having a power saving mechanism.

2. Description of Related Art

(2) An electronic apparatus uses a processing circuit thereof to access commands and data from a memory connected thereto, to perform operation on the commands and the data so as to obtain the processed result in need. In recent years, the power-saving demands on the electronic products become higher and higher. A more efficient access method is therefore needed when the processing circuit accesses the commands from the memory in order to save power.

SUMMARY OF THE INVENTION

(3) In consideration of the problem of the prior art, an object of the present disclosure is to provide an electronic apparatus and a command accessing method thereof having a power saving mechanism.

(4) The present invention discloses a command accessing method having a power saving mechanism that includes steps outlined below. A cache control circuit is controlled to retrieve a plurality of commands from a command source to be executed. A plurality of loop commands that correspond to a loop procedure comprised in the commands are determined, wherein the loop commands comprise a first part having a size that matches a storage amount of a command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit. The cache control circuit is controlled to store the first part of the loop commands to the command buffer circuit and not store the second part of the loop commands to the command buffer circuit. Every time the loop procedure is performed, the cache control circuit is controlled to retrieve and execute the first part of the loop commands from the command buffer circuit and retrieve and execute the second part of the loop commands from the command source.

(5) The present invention also discloses an electronic apparatus having a power saving mechanism that includes a cache control circuit, a command buffer circuit and a processing circuit. The processing circuit is configured to perform steps outlined below. The cache control circuit is controlled to retrieve a plurality of commands from a command source to be executed. A plurality of loop commands that correspond to a loop procedure comprised in the commands are determined, wherein the loop commands comprise a first part having a size that matches a storage amount of the command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit. The cache control circuit is controlled to store the first part of the loop commands to the command buffer circuit and not store the second part of the loop commands to the command buffer circuit. Every time the loop procedure is performed, the cache control circuit is controlled to retrieve and execute the first part of the loop commands from the command buffer circuit and retrieve and execute the second part of the loop commands from the command source.

(6) These and other objectives of the present disclosure will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiments that are illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a block diagram of an electronic apparatus having a power saving mechanism according to an embodiment of the present invention.

(2) FIG. 2 illustrates a diagram of the commands that the cache control circuit retrieves according to an embodiment of the present invention.

(3) FIG. 3 illustrates a flow chart of a command accessing method having a power saving

mechanism according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(4) An aspect of the present invention is to provide an electronic apparatus and a command accessing method thereof having a power saving mechanism to categorize loop commands into a first part having a size matching a storage amount of a command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit such that only the first part is stored in the command buffer circuit. In the execution of every loop procedure, the first part of the loop commands is retrieved and executed from the command buffer circuit and the second part of the loop commands are retrieved and executed from a command source to save power.

(5) Reference is now made to FIG. 1. FIG. 1 illustrates a block diagram of an electronic apparatus **100** having a power saving mechanism according to an embodiment of the present invention. The electronic apparatus **100** includes a cache control circuit **110**, a command buffer circuit **120** and a processing circuit **130**.

(6) In an embodiment, the processing circuit **130** is a core of a central processing unit (CPU) and is electrically coupled to the cache control circuit **110** to control the cache control circuit **110** to retrieve and execute commands CM from a command source. The command source can be such as, but not limited to a system bus **140**, a cache circuit **150**, a memory circuit **160** or a combination thereof further included in the electronic apparatus **100**.

(7) The processing circuit **130** determines that a plurality of loop commands that correspond to a loop procedure included in the commands CM, wherein the loop commands include a first part having a size that matches a storage amount of the command buffer circuit **120** and a second part that exceeds the storage amount of the command buffer circuit **120**. More specifically, the storage amount of the command buffer circuit **120** can only store the first part, in which the second part is not able to be stored in the command buffer circuit **120** when the first part is already stored in the command buffer circuit **120**.

(8) Reference is now made to FIG. 2 at the same time. FIG. 2 illustrates a diagram of the commands CM that the cache control circuit **110** retrieves according to an embodiment of the present invention.

(9) In an embodiment, the commands CM that the cache control circuit **110** retrieves include individual commands respectively labeled as IN(1)~IN(n+2) to be executed by the processing circuit **130** in series, in which n is an integer.

(10) The processing circuit **130** that a current command that is being executed in the commands CM includes a branch request corresponding to a previous command, so as to further determine that the commands from the previous command to the current command are the loop commands CL.

(11) For example, when the processing circuit **130** executes the command IN(n) and determines that the command IN(n) includes a branch request corresponding to the command IN(3), the processing circuit **130** further determines that the commands from the command IN(3) to the command IN(n) are the loop commands CL. These loop commands CL correspond to a loop procedure to be executed by the processing circuit **130** for a plurality of times. More specifically, the processing circuit **130** executes the loop procedure for a plurality of times, in which the commands from the command IN(3) to the command IN(n) are executed in series every time the loop procedure is executed.

(12) In an embodiment, the command buffer circuit **120** is a register having a plurality of D flip-flops configured to store at least a part of the loop commands CL. Since the storage amount of the command buffer circuit **120** is limited, not all the loop commands CL can be stored therein when the amount of the loop commands CL is larger than the storage amount of the command buffer circuit **120**.

(13) As a result, the processing circuit **130** calculates a branch jump distance from the previous command to the current command, in which the branch jump distance equals to the amount of these

loop commands CL. Take the condition described above as an example, the processing circuit **130** calculates the branch jump distance from the command IN(3) to the command IN(n) to be $n-3+1=n-2$.

(14) Moreover, the processing circuit **130** controls the cache control circuit **110** to determine a first part P1 and a second part P2 of the loop commands CL according to the branch jump distance and the storage amount of the command buffer circuit **120**. The first part P1 is a part of the commands in the loop commands CL has a size that matches the storage amount of the command buffer circuit **120**. The second part P2 is the other part of the commands in the loop commands CL that exceeds the storage amount of the command buffer circuit **120**.

(15) Take FIG. 2 as an example, when the storage amount of the command buffer circuit **120** only allows an amount of m loop commands to be stored, the commands from the command IN(3) to the command IN(m+2) correspond to the first part P1 of the loop commands CL. The commands from the command IN(m+3) to the command IN(n) correspond to the second part P2 of the loop commands CL, in which m is a positive integer smaller than n.

(16) The processing circuit **130** controls the cache control circuit **110** to store the first part P1 of the loop commands CL to the command buffer circuit **120**, and not store the second part P2 of the loop commands CL to the command buffer circuit **120**.

(17) Further, every time the loop procedure is performed, the processing circuit **130** controls the cache control circuit **110** to retrieve and execute the first part P1 of the loop commands CL from the command buffer circuit **120**, and retrieve and execute the second part P2 of the loop commands CL from the command source.

(18) More specifically, when the processing circuit **130** executes the commands from the command IN(3) to the command IN(n), only the first time is executed by controlling the cache control circuit **110** to retrieve the commands from the command IN(3) to the command IN(n) directly from the command source (e.g., the system bus **140**, the cache circuit **150**, the memory circuit **160** or a combination thereof). Subsequently, every time the loop procedure is executed, the processing circuit **130** controls the cache control circuit **110** to retrieve and execute the commands from the command IN(3) to the command IN(m+2) (corresponding to the first part P1 of the loop commands CL) from the command buffer circuit **120** instead of the original command source. Further, the processing circuit **130** controls the cache control circuit **110** to retrieve and execute the commands from the command IN(m+3) to the command IN(n) (corresponding to the second part P2 of the loop commands CL) from the command source.

(19) It is appreciated that when the amount of the loop commands CL is smaller than the storage amount of the command buffer circuit **120**, the command buffer circuit **120** is able to store all the loop commands CL. As a result, the loop commands CL only includes the first part P1, and the second part P2 does not exist. Under such a condition, the processing circuit **130** stores all the loop commands CL in the command buffer circuit **120**, such that every time the loop procedure is executed, the processing circuit **130** controls the cache control circuit **110** to retrieve and execute the loop commands CL from the command buffer circuit **120**.

(20) The original object of the command buffer circuit is to store the loop commands such that the processing circuit can quickly access the commands in the loop procedure from the command buffer circuit instead of the original command source so as to save power. However, in some approaches, the processing circuit controls the cache control circuit to directly store all the loop commands in the command buffer circuit. When the storage amount of the command buffer circuit is not able to store all the commands, the cache control circuit keeps overwriting the loop commands that are already stored in the command buffer circuit with the subsequently commands. When the loops procedure is executed again, the cache control circuit is not able to access the correct loop commands from the command buffer circuit and has to access the loop commands from the original command source. Such a strategy cannot save power.

(21) As a result, the electronic apparatus having the power saving mechanism of the present

invention categorizes loop commands into a first part having a size matching a storage amount of a command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit such that only the first part is stored in the command buffer circuit. In the execution of every loop procedure, the first part of the loop commands is retrieved and executed from the command buffer circuit and the second part of the loop commands are retrieved and executed from a command source to save power.

(22) Reference is now made to FIG. 3. FIG. 3 illustrates a flow chart of a command accessing method **300** having a power saving mechanism according to an embodiment of the present invention.

(23) Besides the apparatus described above, the present invention further discloses the command accessing method **300** that can be used in such as, but not limited to the electronic apparatus **100** illustrated in FIG. 1. An embodiment of the command accessing method **300** is illustrated in FIG. 3 and includes the steps outlined below.

(24) In step **S310**, the processing circuit **130** controls the cache control circuit **110** to retrieve the commands **CM** from the command source to be executed.

(25) In step **S320**, the processing circuit **130** determines the loop commands **CL** that correspond to the loop procedure included in the commands **CM**, wherein the loop commands **CL** include the first part **P1** having the size that matches the storage amount of the command buffer circuit **120** and the second part **P2** that exceeds the storage amount of the command buffer circuit **120**.

(26) In step **S330**, the processing circuit **130** controls the cache control circuit **110** to store the first part **P1** of the loop commands **CL** to the command buffer circuit **120** and not store the second part **P2** of the loop commands **CL** to the command buffer circuit **120**.

(27) In step **S340**, every time the loop procedure is performed, the processing circuit **130** controls the cache control circuit **110** to retrieve and execute the first part **P1** of the loop commands **CL** from the command buffer circuit **120** and retrieve and execute the second part **P2** of the loop commands **CL** from the command source.

(28) It is appreciated that the embodiments described above are merely an example. In other embodiments, it is appreciated that many modifications and changes may be made by those of ordinary skill in the art without departing, from the spirit of the invention.

(29) In summary, the electronic apparatus and the command accessing method thereof having the power saving mechanism categorize loop commands into a first part having a size matching a storage amount of a command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit such that only the first part is stored in the command buffer circuit. In the execution of every loop procedure, the first part of the loop commands is retrieved and executed from the command buffer circuit and the second part of the loop commands are retrieved and executed from a command source to save power.

(30) The aforementioned descriptions represent merely the preferred embodiments of the present disclosure, without any intention to limit the scope of the present disclosure thereto. Various equivalent changes, alterations, or modifications based on the claims of present disclosure are all consequently viewed as being embraced by the scope of the present disclosure.

Claims

1. A command accessing method having a power saving mechanism, comprising: controlling a cache control circuit to retrieve a plurality of commands; determining a plurality of loop commands that correspond to a loop procedure comprised in the commands, wherein the loop commands comprise a first part having a size that matches a storage amount of a command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit; controlling the cache control circuit to store the first part of the loop commands to the command buffer circuit and not store the second part of the loop commands to the command buffer circuit; and every time the loop

- procedure is performed, controlling the cache control circuit to retrieve and execute the first part of the loop commands from the command buffer circuit and retrieve and execute the second part of the loop commands.
2. The command accessing method of claim 1, wherein the command buffer circuit is a register having a plurality of D flip-flops.
 3. The command accessing method of claim 1, further comprising: determining that a current command that is being executed in the commands comprises a branch request corresponding to a previous command, so as to further determine that the commands from the previous command to the current command are the loop commands.
 4. The command accessing method of claim 3, further comprising: calculating a branch jump distance from the previous command to the current command; and controlling the cache control circuit to determine the first part and the second part of the loop commands according to the branch jump distance and the storage amount of the command buffer circuit.
 5. An electronic apparatus having a power saving mechanism, comprising: a cache control circuit; a command buffer circuit; and a processing circuit configured to: control the cache control circuit to retrieve a plurality of commands determine a plurality of loop commands that correspond to a loop procedure comprised in the commands, wherein the loop commands comprise a first part having a size that matches a storage amount of the command buffer circuit and a second part that exceeds the storage amount of the command buffer circuit; control the cache control circuit to store the first part of the loop commands to the command buffer circuit and not store the second part of the loop commands to the command buffer circuit; and every time the loop procedure is performed, control the cache control circuit to retrieve and execute the first part of the loop commands from the command buffer circuit and retrieve and execute the second part of the loop commands.
 6. The electronic apparatus of claim 5, wherein the command buffer circuit is a register having a plurality of D flip-flops.
 7. The electronic apparatus of claim 5, wherein the processing circuit is further configured to: determine that a current command that is being executed in the commands comprises a branch request corresponding to a previous command, so as to further determine that the commands from the previous command to the current command are the loop commands.
 8. The electronic apparatus of claim 7, wherein the processing circuit is further configured to: calculate a branch jump distance from the previous command to the current command; and control the cache control circuit to determine the first part and the second part of the loop commands according to the branch jump distance and the storage amount of the command buffer circuit.
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